

SQL Assessment: Employees & Sales Database

You will complete three tasks using the `people1` database. Follow the instructions carefully.

TASK 1 – Query Challenges (15–20 minutes)

Answer the following questions using SQL queries:

1. List the first name, last name, and salary of all employees who earn more than the average salary in the company.
 2. Show the total salary per department, but only include departments where the total salary exceeds 60,000.
 3. Count how many employees are in each department.
 4. Show all employees whose last names contain the letter “e” AND whose salary is below 4000.
 5. Identify the highest salary in the “Private Individuals” department.
 6. Identify the lowest salary in the “Corporate” department.
 7. List all employees whose first name starts with ‘A’ OR ends with ‘a’.
 8. Show all employees sorted by salary in descending order. Display only first name, last name, and salary.
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TASK 2 – Create a New Table + JOIN (15 minutes)

Part A – Departments Table

Create a new table called `departments` with columns for department ID and department name. Then insert two departments: Corporate and Private Individuals.

Part B – Foreign Key

Add a new column `department_id` to the employees table and set it as a foreign key referencing the departments table.

Part C – Update Employees

Update all employees so they have the correct department ID.

Part D – JOIN Queries

Answer the following using JOINS: 1. Show first name, last name, and department name for all employees. 2. Show the total salary per department using the departments table. 3. List all employees in the Corporate department using a JOIN. 4. Show the average salary per department, rounded to two decimal places.

TASK 3 – Company Salary Report (10–15 minutes)

Create a series of SQL queries to produce a company salary report:

1. Show the number of employees in each department.
 2. Show the total salary cost per department.
 3. Show the highest and lowest salary in each department.
 4. Show the average salary in each department, rounded to two decimal places.
 5. Produce a final report listing all employees sorted by department (A–Z) and salary (highest to lowest).
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Optional Extension (if you finish early)

- Identify employees whose salaries are more than one standard deviation above the company average.
- Create a salary bands table and classify employees as Low, Medium, High, or Very High.

Focus on completing the tasks independently. Work carefully, and check your results for accuracy.